A. Discuss Briefly

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

An algorithm is a set of clear, step-by-step instructions used to solve a problem or complete a task. In daily life, we use algorithms even without realizing it. For example, when preparing food, we follow steps such as gathering ingredients, preparing them, cooking them, and serving the food. Following the steps in the correct order helps us complete the task properly.

2. Why is it important for an IT student to learn algorithms?

Algorithms are important for IT students because they help develop logical and problem-solving skills. Programming requires giving a computer precise instructions, so understanding algorithms makes it easier to create programs, find errors, and improve the efficiency of code. Algorithms are also the foundation of many programming concepts and computer applications.

B. Convert the JavaScript Code into Step-by-Step Algorithms

1. First JavaScript Code

Algorithm:
1. Start with the value of i equal to 1.
2. Check if it is less than 10.
3. If it is less than 10, display the value of i.
4. Multiply it by 2.
5. Check again if it is less than 10.
6. Continue displaying and multiplying i by 2 while i is less than 10.
7. When it becomes 16, the condition i < 10 is false.
8. Stop the loop.

2. Second JavaScript Code

Algorithm:
1. Start with the value of i equal to 0.
2. Check if it is less than 10.
3. If it is less than 10, display the value of i.
4. Increase i by 1.
5. Check the condition again.
6. Continue displaying the number and increasing it by 1 while i is less than 10.
7. When i becomes 10, the condition i < 10 is false.
8. Stop the loop.

C. Observations

1. What did you observe about the sequence of numbers produced by the first code?

The first code produces 1, 2, 4, and 8. Each number is twice the previous number. The loop stops after 8 because the next value would be 16, which is no longer less than 10.

2. What did you observe about the sequence of numbers produced by the second code?

The second code produces numbers from 0 to 9 in order. Each number increases by exactly 1 until the value reaches 10, at which point the loop stops.

3. What is the main difference between the way the two loops progress?

The main difference is how the loop variable changes:

- First loop: i *= 2 means the value is multiplied by 2 each time: 1 → 2 → 4 → 8 → 16.
- Second loop: i++ means the value is increased by 1 each time: 0 → 1 → 2 → 3 → ... → 10.

Therefore, the first loop progresses by doubling, while the second progresses by adding one.

4. How does the way an algorithm performs its steps affect its output?

The way an algorithm performs its steps directly affects its output. Even though both programs use a for loop and have a condition involving 10, they produce different sequences because the loop variables are updated differently. This shows that the order and rules of an algorithm determine how a program behaves and what result it produces.

Jimuel D. Asico
BSIT WMT 2-2